

TO: SwiftCart Senior Leadership
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DATE: June 2025
SUBJECT: Week 5 — Localized Forecasting Architecture: Performance Report

1. Overview

This memo evaluates transitioning SwiftCart's demand forecasting from a centralized model to localized forecasting across 50 dark stores. Two localization strategies were compared with a global baseline using **WAPE** and **Total Business Cost**. Training used **Jan–Sep 2023 (53.2k rows)** and testing used **Oct–Dec 2023 (18.4k rows)**.

2. Architecture Comparison Table

Architecture	WAPE (%)	Total Business Cost	vs Baseline
Centralized Global (no store features)	25.20%	\$2,613,815	—
Global + Store Features (target encoded)	25.28%	\$2,615,100	−0.3%
50 Micro-Models (many-model pipeline)	26.22%	\$2,720,942	−4.0%

The centralized global model outperformed both localization strategies on this dataset.

3. Strategy Explanations

Global + Store Features: A single LightGBM model using target-encoded **Store_ID** and **Store_Type** produced virtually no improvement (−0.3%), indicating limited value from encoded store identifiers alone.

asd50 Micro-Models: Separate LightGBM models (~1,064 rows/store) with SKU-level target encoding increased WAPE by 4.0% and outperformed the baseline in only **2/50 stores**, missing the **40-store** validation target.

4. Localization Breakdown — Best and Worst Stores

Best-performing stores (lowest):

Store	Baseline WAPE	Micro-Model WAPE
ST-039	23.29%	23.56%
ST-008	22.60%	23.68%
ST-026	23.25%	23.81%
ST-037	23.20%	24.16%
ST-049	23.49%	24.53%

Worst-performing stores (highest):

Store	Baseline WAPE	Micro-Model WAPE
ST-032	27.05%	28.01%
ST-022	26.21%	28.30%
ST-042	26.71%	28.95%+

Pattern driving this discrepancy: Observation: Stores with low baseline WAPE remained the most predictable, while high-variance stores suffered from overfitting due to limited per-store data. The global model generalized better by leveraging the full dataset (53.2k rows).

5. Conclusion & Recommendation

Localization did not outperform the global baseline. The **12% WAPE improvement target** and **40/50 store validation threshold** were not met.

Recommendation: Deploy the global model and consider a hybrid approach, fine-tuning only stores with sufficient historical data while incorporating external signals (e.g., local events and weather).